



A1	ACE	☒	☐
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Find the primitive function of $2x+9$.

1

$$\int 2x+9dx = x^2+9x+c$$

A2	ARC	☒	☐
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Find the primitive of x^3+2 .

1

$$\frac{x^4}{4}+2x+c$$

A3	ARC	☒	☐
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Find $\int (4x+3)^2 dx$.

1

$$\int (4x+3)^2 dx = \frac{(4x+3)^3}{12} + c$$

A4	Ascham	☒	☐
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Find $\int \frac{1}{x^2} dx$

(A) $\log(x^2)+c$

(C) $\frac{-1}{x}+c$

(B) $\frac{-2}{x^3}+c$

(D) $2x\log(x^2)+c$

A5	Barker	☒	☐
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Find $\int \frac{dx}{\sqrt{x^2}}$.

2

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2}} &= \int \frac{dx}{x^{\frac{1}{2}}} \\ &= \int x^{-\frac{1}{2}} dx \\ &= \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + c \\ &= \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c \\ &= 2\sqrt{x} + c \end{aligned}$$

A6	Barker	☒	☐
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$$\int_0^1 3\sqrt{x} dx$$

$$\begin{aligned} \therefore \int_0^1 3\sqrt{x} dx &= \left[\frac{3x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 \\ &= \left[2x^{\frac{3}{2}} \right]_0^1 \\ &= 2-0 \\ &= 2 \end{aligned}$$

2

A7	ACE	☒	☐
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A primitive function of $4+\sqrt{x}$ is

(A) $4x + \frac{2\sqrt{x^3}}{3}$

(B) $4x + \frac{3\sqrt{x^2}}{2}$

(C) $4x + \frac{1}{2\sqrt{x}}$

(D) $\frac{1}{2\sqrt{x}}$

$$\begin{aligned} y' &= 4 + \sqrt{x} \\ &= 4 + x^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} y &= 4x + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \\ &= 4x + \frac{4\sqrt{x^3}}{3} \end{aligned}$$

(A)

A8	ACE	☒	☐
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$$\int \frac{1-x^2}{x} dx$$

$$\begin{aligned} \int \frac{1-x^2}{x} dx &= \int \frac{1}{x} - x dx \\ &= \ln x - \frac{x^2}{2} + c \end{aligned}$$

2

A9	Baulkham Hills	☒	☐
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$$\int_{-2}^2 3x-5 dx =$$

(A) 10

(B) -10

(C) 20

(D) -20



A10	Caringbah	☒	☐
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Find $\int (2x-5)^3 dx$

$$\int (2x-5)^3 dx = \frac{2x-5^4}{8} + C$$

1

A11	Caringbah	☒	☐
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Find $\int (3x-2)^5 dx$.

$$\int (3x-2)^5 dx = \frac{(3x-2)^6}{18} + C$$

1

A12	CSSA	☒	☐
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Find the value of k given $\int_0^2 4x+k dx = 18$.

Sample answer

$$\int_0^2 4x+k dx = [2x^2+kx]_0^2$$

$$\therefore 8+2k=18$$

$$k=5$$

2

A13	Gosford	☒	☐
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Find $\int \frac{x^3+x}{2x^2} dx$

$$\int \frac{x^3}{2x^2} + \frac{x}{2x^2} dx$$

$$= \int \frac{x}{2} + \frac{1}{2x} dx$$

$$= \frac{x^2}{4} + \frac{1}{2} \ln x + C$$

2

A14	Girraween	☒	☐
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Find $\int \frac{2}{(x-5)^2} dx$

$$\int \frac{2}{(x-5)^2} dx$$

$$= 2 \int (x-5)^{-2} dx$$

$$= -2(x-5)^{-1} + C$$

$$= \frac{-2}{x-5} + C \quad \#$$

2

A15	Girraween	☒	☐
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Evaluate $\int_0^3 \sqrt{5x+1} dx$

$$\int_0^3 \sqrt{5x+1} dx$$

$$= \left[\frac{(5x+1)^{3/2}}{5 \times 3/2} \right]_0^3$$

$$= \left[\frac{2}{15} (5x+1)^{3/2} \right]_0^3$$

$$= \frac{2}{15} \times 64 - \frac{2}{15}$$

$$= \frac{2}{15} \times 16^{3/2} - \frac{2}{15} \times 1^{3/2} = \frac{2}{15} \times 64 - \frac{2}{15} = \frac{2}{15} \times 63 = \frac{2}{5} \times 9 = \frac{18}{5} \quad \textcircled{3}$$

2

A16	Caringbah	☒	☐
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Evaluate $\int_1^2 \frac{1}{(3x-1)^2} dx$

$$\int_1^2 \frac{1}{(3x-1)^2} dx = \int_1^2 (3x-1)^{-2} dx$$

$$= \left[\frac{(3x-1)^{-1}}{3 \times (-1)} \right]_1^2$$

$$= \left[\frac{-1}{3(3x-1)} \right]_1^2$$

$$= \frac{-1}{15} - \left(\frac{-1}{6} \right)$$

$$= \frac{-1}{15} + \frac{1}{6}$$

$$= \frac{1}{10} \quad \textcircled{3m}$$

2

A17	Kings	☒	☐
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Evaluate $\int \frac{4x^5+1}{4x^4} dx$

$$\int \frac{4x^5+1}{4x^4} dx = \int \frac{4x^5}{4x^4} + \frac{1}{4x^4} dx$$

$$= \int x + \frac{1}{4} x^{-4} dx$$

$$= \frac{x^2}{2} + \frac{1}{4} \cdot \frac{x^{-3}}{-3} + C = \frac{x^2}{2} - \frac{1}{12x^3} + C$$

2

A18	Sydney Tech	☒	☐
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Evaluate $\int_3^8 \sqrt{x+1} dx$

$$\int_3^8 (x+1)^{1/2} dx$$

$$= \left[\frac{2}{3} (x+1)^{3/2} \right]_3^8$$

$$= \frac{2}{3} \left[9^{3/2} - 4^{3/2} \right]$$

$$= \frac{2}{3} [27 - 8]$$

$$= 12^{2/3} \quad \left(\frac{38}{3} \right)$$

A19	Girraween	☒	☐
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Evaluate $\int_0^9 (\sqrt{x}+1)^2 dx$

$$a) \int_0^9 (\sqrt{x}+1)^2 dx$$

$$= \int_0^9 (x+2x^{1/2}+1) dx$$

$$= \left[\frac{x^2}{2} + \frac{4x^{3/2}}{3} + x \right]_0^9$$

$$= \left(\frac{81}{2} + 36 + 9 \right) - (0) = 85 \frac{1}{2}$$

A20	Girraween	☒	☐
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Find the value of a if $\int_2^a (2x+1) dx = 14$

$$\int_2^a (2x+1) dx = \left[x^2 + x \right]_2^a$$

$$= a^2 + a - 6$$

$$a^2 + a - 20 = 0$$

$$(a+5)(a-4) = 0$$

$$a = 4, -5$$

A21	Blacktown Boys	☒	☐
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Evaluate $\int_0^2 \frac{3}{7x+2} dx$

$$\int_0^2 \frac{3}{7x+2} dx$$

$$= \frac{3}{7} \ln|7x+2| \Big|_0^2$$

$$= \frac{3}{7} (\ln 14 - \ln 2)$$

$$= \frac{3}{7} \ln 7$$

A22	Gosford	☒	☐
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Find the values of any constant A such that $\int_0^3 (x^2 + A) dx = 2A^2$

$$\int_0^3 (x^2 + A) dx = 2A^2$$

$$\left[\frac{x^3}{3} + Ax \right]_0^3 = 2A^2$$

$$\frac{27}{3} + 3A = 2A^2$$

$$9 + 3A = 2A^2$$

$$2A^2 - 3A - 9 = 0$$

$$2A^2 - 6A + 3A - 9 = 0$$

$$2A(A-3) + 3(A-3) = 0$$

$$(A-3)(2A+3) = 0$$

$$A = 3 \text{ or } A = -\frac{3}{2}$$

A23	Hurlstone	☒	☐
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What is the value of $\int_{-4}^4 \sqrt{16-x^2} dx$?

A. 16

B. 4π

C. 8π

D. 32

This is the top semi-circle with radius 4.

$$\int_{-4}^4 \sqrt{16-x^2} dx = \frac{1}{2} \times \pi \times 4^2 = 8\pi$$

$\therefore C$



A24	Penrith	☒	☐
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Evaluate $\int_2^4 \frac{2x^2 - 1}{x} dx$

$$= \int_2^4 \left(\frac{2x^2}{x} - \frac{1}{x} \right) dx$$

$$= \int_2^4 \left(2x - \frac{1}{x} \right) dx$$

$$= \left[x^2 - \ln|x| \right]_2^4$$

$$= 12 - \ln 4 + \ln 2$$

$$= 12 - 2\ln 2 + \ln 2$$

$$= 12 - \ln 2$$

A25	St George Girls	☒	☐
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What is $\int \frac{4}{(5-2x)^3} dx$?

(A) $\frac{1}{(5-2x)^2} + C$

(B) $\frac{1}{2(5-2x)^2} + C$

(C) $\frac{-1}{2(5-2x)^2} + C$

(D) $\frac{-1}{(5-2x)^2} + C$

$$\int \frac{4}{(5-2x)^3} dx$$

$$= \int 4(5-2x)^{-3} dx$$

$$= -2 \int -2(5-2x)^{-3} dx$$

$$= -2 \cdot \frac{(5-2x)^{-2}}{-2} + C$$

$$= \frac{1}{(5-2x)^2} + C$$

A26	St George Girls	☒	☐
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Evaluate $\int \frac{x^2 - 3x + 4}{\sqrt{x}} dx$

a) $\int \frac{x^2 - 3x + 4}{\sqrt{x}} dx$

$$= \int \frac{x^2}{x^{\frac{1}{2}}} - \frac{3x}{x^{\frac{1}{2}}} + \frac{4}{x^{\frac{1}{2}}} dx$$

$$= \int x^{\frac{3}{2}} - 3x^{\frac{1}{2}} + 4x^{-\frac{1}{2}} dx$$

$$= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} - \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{4x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= \frac{2}{5}x^{\frac{5}{2}} - 2x^{\frac{3}{2}} + 8x^{\frac{1}{2}} + C$$

A27	Sydney Girls	☒	☐
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Evaluate $\int_1^{\sqrt{3}} \left(\frac{x^3}{6} + \frac{1}{3x^2} \right) dx$, giving your answer in the form $a + b\sqrt{3}$, where a and b are constants to be determined.

$$= \left[\frac{x^4}{24} - \frac{1}{3x} \right]_1^{\sqrt{3}}$$

$$= \frac{(\sqrt{3})^4}{24} - \frac{1}{3\sqrt{3}} - \left(\frac{1^4}{24} - \frac{1}{3} \right)$$

$$= \frac{1}{24} - \frac{\sqrt{3}}{9} + \frac{7}{24}$$

$$= \frac{2}{3} - \frac{\sqrt{3}}{9}$$

A28	Sydney Grammar	☒	☐
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Evaluate $\int (1-2x)^5 dx$

$$\int (1-2x)^5 dx = \frac{(1-2x)^6}{6 \times -2} = -\frac{1}{12}(1-2x)^6 + C$$

A29	Amity	☒	☐
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(a) Differentiate $\frac{x^2}{x^2+1}$.

(a) $u = x^2, v = x^2+1$

$$\frac{u'v - uv'}{v^2} = \frac{2x(x^2+1) - x^2 \cdot 2x}{(x^2+1)^2}$$

$$= \frac{2x^3 + 2x - 2x^3}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}$$

(b) Hence, evaluate $\int_1^2 \frac{x}{(x^2+1)^2} dx$.

(b) $\int_1^2 \frac{2x}{(x^2+1)^2} dx$

$$= \frac{1}{2} \int_1^2 \frac{2x}{(x^2+1)^2} dx$$

$$= \frac{1}{2} \left[\frac{x^2}{x^2+1} \right]_1^2$$

$$= \frac{1}{2} \left[\frac{4}{5} - \frac{1}{2} \right] = \frac{3}{20}$$



B1	James Ruse	☒	☐
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Evaluate $\int_{-1}^1 \frac{-6x^2}{(x^3+2)^2} dx$. 2

$$\int_{-1}^1 \frac{-2(3x^2)}{(x^3+2)^2} dx = \frac{-2}{-1} \left[(x^3+2)^{-1} \right]_{-1}^1$$

$$= 2 \left((1^3+2)^{-1} - ((-1)^3+2)^{-1} \right) = \frac{-4}{3}$$

B2	Merewether	☒	☐
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Find $\int \frac{3x}{4+2x^2} dx$. 2

$$= \frac{3}{4} \int \frac{4x}{4+2x^2} dx$$

$$= \frac{3}{4} \ln(4+2x^2) + C$$

B3	North Sydney Girls	☒	☐
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Find $\int \frac{x}{\sqrt{x^2+1}} dx$. 2

$$\frac{1}{2} \int 2x(x^2+1)^{-1/2} dx = \frac{1}{2} \cdot 2 \cdot \frac{(x^2+1)^{1/2}}{1/2} + C = \sqrt{x^2+1} + C$$

B4	Sydney Boys	☒	☐
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Find $\int \frac{x}{\sqrt{3x^2+5}} dx$. Using the reference sheet $\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + C$ 2

$$\int \frac{x}{\sqrt{3x^2+5}} dx = \frac{1}{6} \left(\frac{1}{\frac{1}{2}} \right) (3x^2+5)^{\frac{1}{2}} + C$$

$$= \int x(3x^2+5)^{-\frac{1}{2}} dx = \frac{1}{3} (3x^2+5)^{\frac{1}{2}} + C$$

$$= \frac{1}{6} \int 6x(3x^2+5)^{-\frac{1}{2}} dx = \frac{1}{3} \sqrt{3x^2+5} + C$$

B5	VCE	☒	☐
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Calculate $\int 4x^2(x^3+3)^7 dx$. 2

$$\frac{4}{3} \int 3x^2(x^3+3)^7 dx = \frac{4}{3} \frac{(x^3+3)^8}{8} + C$$

$$= \frac{1}{6} (x^3+3)^8 + C$$

B6	VCE	☒	☐
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Calculate $\int (2x+1)\sqrt{x^2+x+1} dx$. 2

$$\int (2x+1)\sqrt{x^2+x+1} dx = \frac{2}{3} (x^2+x+1)^{\frac{3}{2}} + C$$

B7	VCE	☒	☐
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Find $\int 8x^7(x^8-9)^6 dx$, with C as the constant of integration. 2

$$\int 8x^7(x^8-9)^6 dx = \frac{(x^8-9)^7}{7} + C$$

B8	VCE	☒	☐
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a Find $\frac{d}{dx}(2x+3)^4$. 1

b Hence find:

i $\int 8(2x+3)^3 dx$ 2

a $\frac{d}{dx}(2x+3)^4$ 2

$$= 4(2x+3)^3 \times 2$$

$$= 8(2x+3)^3$$

b i $\int 8(2x+3)^3 dx$

$$= \int \frac{d}{dx}(2x+3)^4 dx$$

$$= (2x+3)^4 + C$$

b ii $\int 16(2x+3)^3 dx$

$$= 2 \int 8(2x+3)^3 dx$$

$$= 2 \int \frac{d}{dx}(2x+3)^4 dx$$

$$= 2(2x+3)^4 + C$$



B9	VCE	☒	☐
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a Find $\frac{d}{dx}(1 - 2x)^4$. 1

b Hence find:

i $\int -8(1 - 2x)^3 dx$

ii $\int -2(1 - 2x)^3 dx$ 2

4a $\frac{d}{dx}(1 - 2x)^4$ $= 4(1 - 2x)^3 \times -2$ $= -8(1 - 2x)^3$	4b i $\int -8(1 - 2x)^3 dx$ $= \int \frac{d}{dx}(1 - 2x)^4 dx$ $= (1 - 2x)^4 + C$	4b ii $\int -2(1 - 2x)^3 dx$ $= \frac{1}{4} \int -8(1 - 2x)^3 dx$ $= \frac{1}{4} \int \frac{d}{dx}(1 - 2x)^4 dx$ $= \frac{1}{4}(1 - 2x)^4 + C$
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B10	VCE	☒	☐
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a Find $\frac{d}{dx}(4x + 3)^{-1}$. 1

b Hence find:

i $\int -4(4x + 3)^{-2} dx$

ii $\int (4x + 3)^{-2} dx$ 2

5a $\frac{d}{dx}(4x + 3)^{-1}$ $= -1(4x + 3)^{-2} \times 4$ $= -4(4x + 3)^{-2}$	5b i $\int -4(4x + 3)^{-2} dx$ $= \int \frac{d}{dx}(4x + 3)^{-1} dx$ $= (4x + 3)^{-1} + C$	5b ii $\int (4x + 3)^{-2} dx$ $= -\frac{1}{4} \int -4(4x + 3)^{-2} dx$ $= -\frac{1}{4}(4x + 3)^{-1} + C$
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B11	VCE	☒	☐
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a Find $\frac{d}{dx}(\sqrt{x} + 1)^3$. 1

b Hence find:

i $\int \frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}} dx$

ii $\int \frac{(\sqrt{x} + 1)^2}{\sqrt{x}} dx$ 2

10a $\frac{d}{dx}(\sqrt{x} + 1)^3$ $= 3(\sqrt{x} + 1)^2 \times \frac{1}{2}x^{-\frac{1}{2}}$ $= \frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}}$	10b i $\int \frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}} dx$ $= \int \frac{d}{dx}(\sqrt{x} + 1)^3 dx$ $= (\sqrt{x} + 1)^3 + C$	10b ii $\int \frac{(\sqrt{x} + 1)^2}{\sqrt{x}} dx$ $= \frac{2}{3} \int \frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}} dx$ $= \frac{2}{3} \int \frac{d}{dx}(\sqrt{x} + 1)^3 dx$ $= \frac{2}{3}(\sqrt{x} + 1)^3 + C$
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B12	VCE	☒	☐
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Evaluate these definite integrals using the reverse chain rule.

a $\int_0^1 \frac{x}{(5x^2 + 1)^3} dx$

b $\int_0^{\frac{1}{2}} x\sqrt{1 - 4x^2} dx$ 4

a $\int_0^1 \frac{x}{(5x^2 + 1)^3} dx$

$$= \int_0^1 \frac{1}{(5x^2 + 1)^3} \times x dx$$

$$= \int_1^6 \frac{1}{u^3} \times \frac{1}{10} du$$

$$= \frac{1}{10} \int_1^6 u^{-3} du$$

$$= \frac{1}{10} \left[\frac{u^{-2}}{-2} \right]_1^6 = \frac{1}{10} \left[\frac{1}{-2u^2} \right]_1^6 = \frac{7}{144}$$

b $\int_0^{\frac{1}{2}} x\sqrt{1 - 4x^2} dx$

$$= \int_0^{\frac{1}{2}} \sqrt{1 - 4x^2} \times x dx$$

$$= \int_1^0 \sqrt{u} \times -\frac{1}{8} du$$

$$= -\frac{1}{8} \int_1^0 u^{\frac{1}{2}} du$$

$$= -\frac{1}{8} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^0 = \frac{1}{12}$$

C1	Girraween	☒	☐
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The table below shows the values of a function $f(x) = \sqrt{25 - x^2}$ for six values of x .

x	0	1	2	3	4	5
$f(x)$	5.00	4.90	4.58	4.00	3.00	0.00

What value is an estimate for $\int_0^5 \sqrt{25 - x^2} dx$ using trapezoidal rule with these six values?

- (A) 10.74
(B) 12.65
(C) 18.98
(D) 37.96

C2	Ascham	☒	☐
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- (i) Use the trapezoidal rule with three function values to estimate $\int_0^2 \sqrt{4 - x^2} dx$ (to 2 decimal places) 3
- (ii) Is this approximation an over or under estimate of the exact value? Explain. 1
- (iii) What is the exact value of the definite integral? 2

$$1) \int_0^2 \sqrt{4 - x^2} dx$$

$$\approx \frac{h}{2} [y_0 + y_2 + 2y_1]$$

x	0	1	2
$\sqrt{4 - x^2}$	2	$\sqrt{3}$	0
	y_0	y_1	y_2

$$\approx \frac{1}{2} (2 + 0 + 2\sqrt{3})$$

$$= 1 + \sqrt{3}$$

$$= 2.73$$

ii) Under exact value
because curve concave down
so trapezoid less in area.

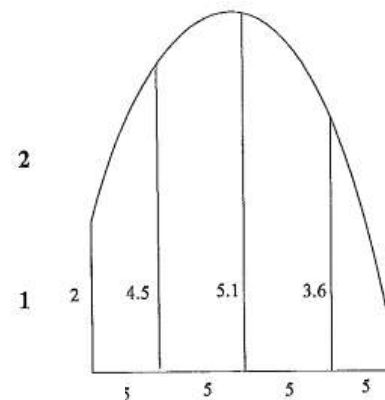
iii) Exact value $= \frac{1}{4} \pi r^2$
 $= \frac{1}{4} \pi \times 2^2$
 $= \pi$

C3	Abbotsleigh	☒	☐
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The diagram below shows a native garden.
All measurements are in metres.

- (i) Use the Trapezoidal Rule with 4 intervals to find an approximate value for the area of the native garden. 2

- (ii) If 25 millimetres of rain fell overnight, how many litres of rain fell on the native garden. Assume $1 \text{ m}^3 = 1000 \text{ litres}$



$$(i) A = \frac{h}{2} (d_f + 2d_m + d_l) + \frac{h}{2} (d_f + 2d_m + d_l)$$

$$= \frac{5}{2} (2 + 2 \times 4.5 + 5.1) + \frac{5}{2} (5.1 + 2 \times 3.6 + 0)$$

$$= 71 \text{ m}^2$$

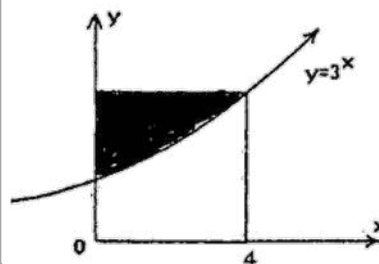
(ii) Now 25 mm = 0.025 m $V = Ah$

$$= 71 \times 0.025$$

$$= 1.775 \text{ m}^3 = 1775 \text{ Litres}$$

C4	Fort Street	☒	☐
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- i) Use the Trapezoidal Rule with 2 strips (i.e. 3 function value), to find an approximate value for $\int_0^4 3^x dx$. 2
- ii) Hence find an approximate value for the shaded area in the diagram below. 2



$$\int_0^4 3^x dx \approx \frac{h}{2} [f(x_0) + f(x_2) + 2f(x_1)]$$

$$= \frac{2}{2} [f(0) + f(4) + 2f(2)]$$

$$= [3^0 + 3^4 + 2 \times 3^2]$$

$$= 100$$

ii) Shaded region $\approx$ rectangle $= 100$

$$= 81 \times 4 = 324$$

$$\text{shaded area} \approx 224 \text{ u}^2$$

C5	CEM	☒	☐
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Consider the function defined by $f(x) = \sqrt{1+x^2}$, where $-1 \leq x \leq 4$.

- (i) Copy and complete the table of values in your book. 1
Give answers correct to 2 decimal places.

x	-1	0	1	2	3	4
$f(x)$						

- (ii) Using the Trapezoidal Rule with six function values, 2
approximate the area under the curve $y = f(x)$ for
 $-1 \leq x \leq 4$. (Give your answer correct to one decimal place.)

- (i) Using the calculator, substitute the given x values into $f(x) = \sqrt{1+x^2}$ to find the $y = f(x)$ values.

x	-1	0	1	2	3	4
$f(x)$	1.41	1.00	1.41	2.24	3.16	4.12
	y_1	y_2	y_3	y_4	y_5	y_6

(ii)

$$\text{Area} = \frac{h}{2} [(y_1 + y_6) + 2(y_2 + y_3 + y_4 + y_5)]$$

Note $h=1$

$$= \frac{1}{2} [(1.41 + 4.12) + 2(1.00 + 1.41 + 2.24 + 3.16)]$$

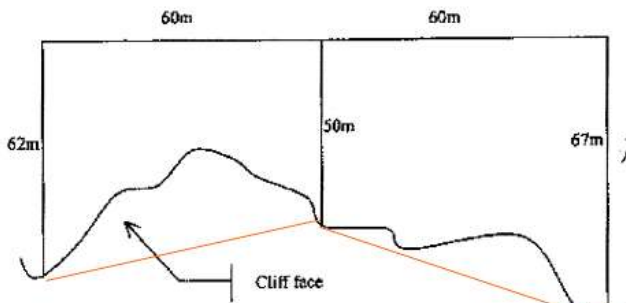
$$= 10.575$$

$$\approx 10.6 \text{ (1dp)}$$

Therefore, the required area is 10.6 unit².

C6	Ascham	☒	☐
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A farmer has a field that is bounded by fences on three straight sides. The fourth boundary is a steep cliff. He needs to determine the area for seed sowing. Distances within the field are given on the diagram. All distances are in metres and the diagram is not to scale.



i) $\text{Area} \approx \frac{1}{2} (y_0 + y_2 + 2y_1)$

$$= \frac{1}{2} (62 + 67 + 2 \times 50)$$

$$= 6870 \text{ m}^2$$

ii) Estimate is greater because the two trapezia formed are greater than the actual area

- (i) Use the trapezoidal rule to determine the approximate area of the field [2]
(ii) Is the estimate greater or less than the true area? Justify your answer. [1]

C7	SCEGGS	☒	☐
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A function is defined as $y = \ln x$.

- (i) Copy the table into your writing book and complete it giving answers to 3 decimal places. 1

x	2	3	4	5
$y = \ln x$				

- (ii) Using the trapezoidal rule, find an approximation, to 1 decimal place, 3
of $\int_2^5 \ln x \, dx$ using 4 function values.
- (iii) Sketch the graph of $y = \ln x$ and use it to explain why your approximation 2
in part (ii) will be less than the exact value of the integral.

b) i)

x	2	3	4	5
$y = \ln x$	0.693	1.099	1.386	1.609

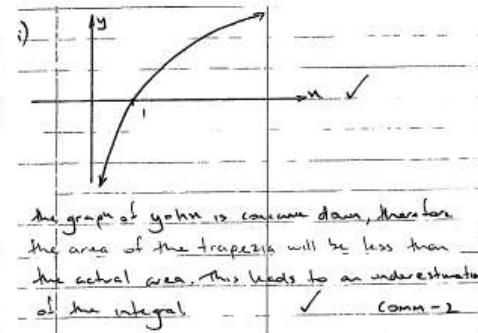
ii)

$$\int_2^5 \ln x \, dx \approx \frac{3-2}{2} [f(2) + f(3)] + \frac{4-3}{2} [f(3) + f(4)] + \frac{5-4}{2} [f(4) + f(5)]$$

$$= \frac{1}{2} [f(2) + 2f(3) + 2f(4) + f(5)]$$

$$= \frac{1}{2} [0.693 + 2 \times 1.099 + 2 \times 1.386 + 1.609]$$

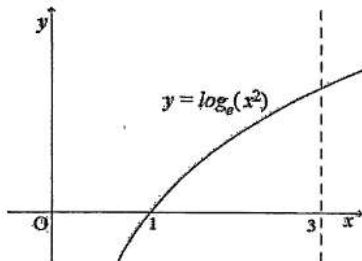
$$= 3.6 \text{ (to 1 d.p.)}$$



C8	Roseville College	☒	☐
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The graph of the function $y = \log_e(x^2)$ is shown below.

3



Use the Trapezoidal rule with 5 function values to approximate $\int_1^3 \log_e(x^2) dx$ and explain that this approximation underestimates the value of the integral.

x	y	w	wy
1	0	1	0
1.5	0.811	2	1.62
2	1.386	2	2.77
2.5	1.833	2	3.67
3	2.197	1	2.197
			$\Sigma wy = 10.25$

$\therefore \int_1^3 \log_e(x^2) dx = \frac{1}{2} \times 2 \times 10.25$
 $= 2.56 (2dp)$
 This underestimates value of the integral as it takes the area of 4 trapeziums under the curve excluding the area between the trapeziums and the curve.

C9	Sydney Girls	☒	☐
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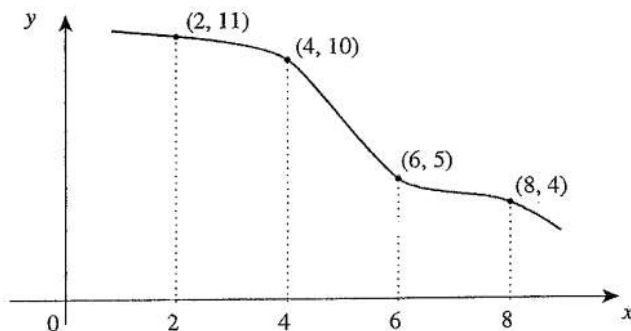
Find an approximation for $\int_2^6 \frac{1}{\sqrt{3x-5}} dx$ correct to 2 decimal places, using 3 applications of the trapezoidal rule.

2

$$\begin{aligned} \text{Area} &= \frac{6-2}{2(3)} \left\{ f(2) + f(6) + 2 \left[f\left(\frac{10}{3}\right) + f\left(\frac{14}{3}\right) \right] \right\} \\ &= \frac{2}{3} \left\{ 1 + \frac{1}{\sqrt{13}} + 2 \left[\frac{1}{\sqrt{5}} + \frac{1}{3} \right] \right\} \\ &\approx 1.89 \end{aligned}$$

C10	NEAP	☒	☐
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2



The diagram shows $y = f(x)$.

Use the trapezoidal rule with four function values (i.e. three applications) to approximate the area enclosed between the curve $y = f(x)$, the x -axis and the lines at $x = 2$ and $x = 8$.

x	2	4	6	8
y	11	10	5	4

$$A \approx \frac{h}{2} \{f + l + 2 \times \text{others}\}$$

$$h = 2$$

$$A \approx \frac{2}{2} \{11 + 4 + 2 \times (10 + 5)\} = 45 \text{ u}^2$$

C11	Caringbah	☒	☐
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Use two applications of the Trapezoidal rule to find an approximate value, correct to 2 decimal places for, $\int_1^3 \ln(x^2+1) dx$.

3

$$\begin{aligned} &\frac{1}{2} \left[\ln(1^2+1) + \ln(3^2+1) + 2 \ln(2^2+1) \right] \\ &= 3.11 \end{aligned}$$

C12	Baulkham Hills	☒	☐
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Use three applications of the trapezoidal rule to estimate the value of $\int_1^4 \sqrt{x-1} \, dx$. Give your answer correct to 4 significant figures. 2

x	1	2	3	4
$f(x)$	0	1	$\sqrt{2}$	$\sqrt{3}$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + f(x_2) + \dots + f(x_{n-1})]\}$$

$$\int_1^4 \sqrt{x-1} dx \approx \frac{4-1}{2 \times 3} \{0 + \sqrt{3} + 2(1 + \sqrt{2})\} \approx 3.280238966 \dots \approx 3.280 \text{ (4 significant figures)}$$

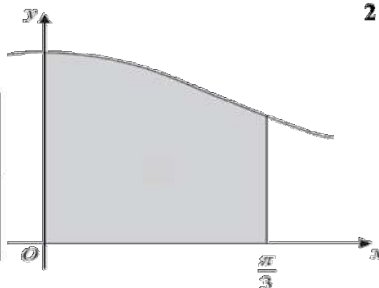
C13	Gosford	☒	☐
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The diagram shows a curve with equation $y = \sqrt{0.75 + \cos^2 x}$. The area is bounded by the curve, the y-axis, the x-axis and the lines with equation $x = 0$ and $x = \frac{\pi}{3}$.

(a) Complete the table with the remaining values of x (evenly spaced) and y for 5 function values.

x	0		$\frac{\pi}{6}$		$\frac{\pi}{3}$
y	1.32		1.22		1

(b) Estimate the area of the shaded region by using the trapezoidal rule, with 5 function values, correct to 2 decimal places. 2



x	0	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
y	1.32	1.29	1.22	1.12	1

$$\begin{aligned} \text{Area} &= \frac{1}{2} \left(\frac{\pi}{12} \right) [1.32 + 2(1.29 + 1.22 + 1.12) + 1] \\ &= 1.26 \text{ u}^2 \end{aligned}$$

C14	Hurlstone	☒	☐
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Given that $y = \log(3^x - 2x)$

a) Complete the table below by finding the missing value of y correct to three decimal places. 1

x	1	1.5	2	2.5	3
y	0	0.787	1.609		3.045

b) Using the table above and the trapezoidal rule, find an approximation for the value of $\int_1^3 \log(3^x - 2x) \, dx$ correct to three decimal places. 2

x	1	1.5	2	2.5	3
y	0	0.787	1.609	2.360	3.045

$$\int_1^3 \log(3^x - 2x) \, dx \approx \frac{3-1}{2(4)} [0 + 3.045 + 2(0.787 + 1.609 + 2.360)] = 3.13925 \approx 3.139$$

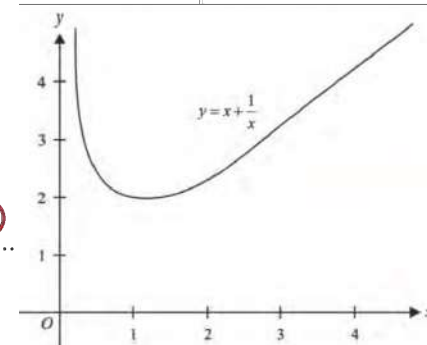
C15	Baulkham Hills	☒	☐
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The graph of $y = x + \frac{1}{x}$ is shown over part of its domain. 2

Use two trapeziums of equal width to approximate the area between the curve, the x-axis and the lines $x = 1$ and $x = 3$

x	1	2	3
y	2	$2\frac{1}{2}$	$3\frac{1}{3}$

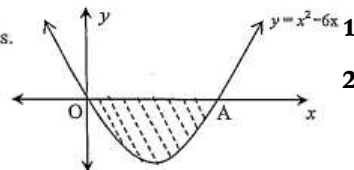
$$\begin{aligned} A &= \frac{b-a}{2} (\text{1st} + 2 \times \text{middles} + \text{last}) \\ &= \frac{1}{2} \left(2 + 2 \times \frac{5}{2} + \frac{10}{3} \right) \\ &= \frac{1}{2} \left(2 + 5 + \frac{10}{3} \right) \\ &= \frac{1}{2} \left(\frac{31}{3} \right) = \frac{31}{6} = 5\frac{1}{6} \end{aligned}$$



D1	Cranbrook	☒	☐
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The above diagram shows the graph of the function $y = x^2 - 6x$.

- Find the coordinates of the point A where the curve crosses the x-axis.
- Find the area of the shaded region contained by the curve and the x-axis.



Question 5 (a) (i)

Answer:

To find where the curve cuts x-axis, put $x^2 - 6x = 0 \Rightarrow x(x-6) = 0$
i.e. $x = 0$ or $x = 6 \therefore A$ is $(6, 0)$ ✓

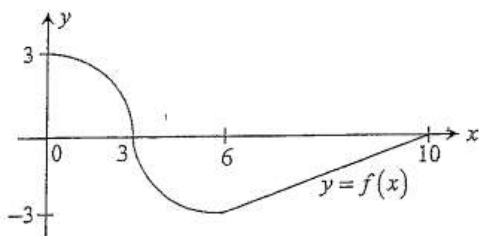
Question 5 (a) (ii)

Answer:

$$\begin{aligned} \text{shaded area} &= \left| \int_0^6 (x^2 - 6x) dx \right| \quad \checkmark \\ &= \left| \left[\frac{x^3}{3} - 3x^2 \right]_0^6 \right| = |-36| = 36 \text{ unit}^2 \quad \checkmark \end{aligned}$$

D2	Abbotsleigh	☒	☐
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The following illustrates the function, $y = f(x)$ for $x = 0$ to $x = 10$.

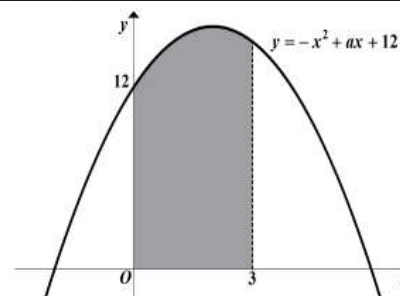


The curve from $x = 0$ to $x = 3$ is a part of a circle of radius 3 units, similarly for that part of the curve from $x = 3$ to $x = 6$. From $x = 6$ to $x = 10$, the function is a straight line.

Explain why $\int_0^{10} f(x) dx = -6$.

$$\begin{aligned} \therefore \int_0^{10} f(x) dx &= \int_0^3 f(x) dx + \int_3^6 f(x) dx + \int_6^{10} f(x) dx \\ &= 0 + (-6) \\ &= -6 \end{aligned}$$

D3	Caringbah	☒	☐
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Part of the graph of the function $y = -x^2 + ax + 12$, is shown below.

If the shaded area is 45 square units, find the values of a .

$$\int_0^3 -x^2 + ax + 12 dx = 45$$

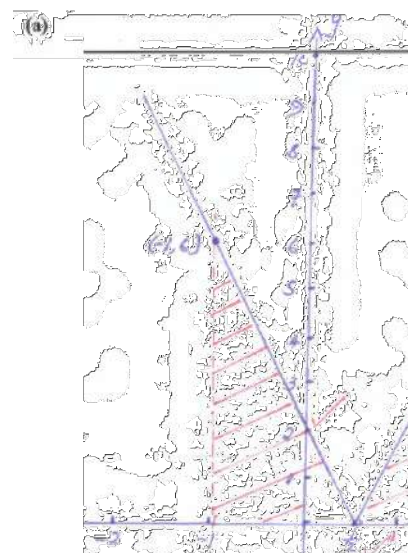
$$\therefore \left[-\frac{x^3}{3} + \frac{ax^2}{2} + 12x \right]_0^3 = 45$$

$$-9 + \frac{9a}{2} + 36 = 45 \rightarrow a = 4$$

D4	Blacktown Boys	☒	☐
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Given the function $y = |4x - 2|$

- Sketch $y = |4x - 2|$, showing all key features. 2
- Hence evaluate $\int_{-1}^3 |4x - 2| dx$. 2

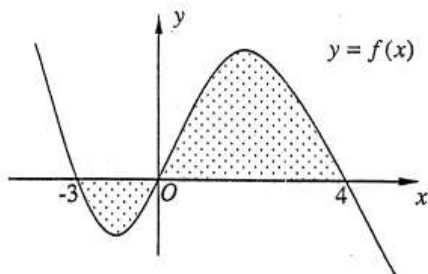


(b)

$$\begin{aligned} &= \int_{-1}^3 |4x - 2| dx \\ &= \int_{-1}^{0.5} (2 - 4x) dx + \int_{0.5}^3 (4x - 2) dx \\ &= \left[2x - 2x^2 \right]_{-1}^{0.5} + \left[2x^2 - 2x \right]_{0.5}^3 \\ &= \left(1 - 0.5 \right) - \left(-2 - 0.5 \right) + \left(18 - 6 \right) - \left(0.5 - 1 \right) \\ &= 1 - 0.5 + 2.5 + 17.5 - 0.5 + 1 \\ &= 21 \end{aligned}$$

D5	CSSA	☒	☐
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Consider the diagram below.



Which of the following represents the shaded area?

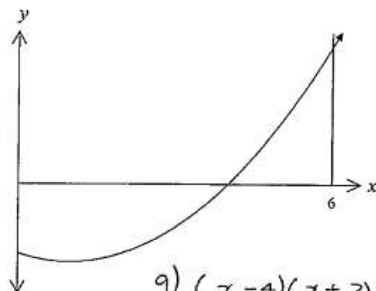
- (A) $\int_{-3}^4 f(x) dx$ (B) $2 \int_0^4 f(x) dx$
- (C) $\int_0^4 f(x) dx - \int_{-3}^0 f(x) dx$ (D) $\int_{-3}^0 f(x) dx + \int_0^4 f(x) dx$

D6	Girraween	☒	☐
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The diagram below shows the graph of $y = x^2 - 2x - 8$.

What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$?

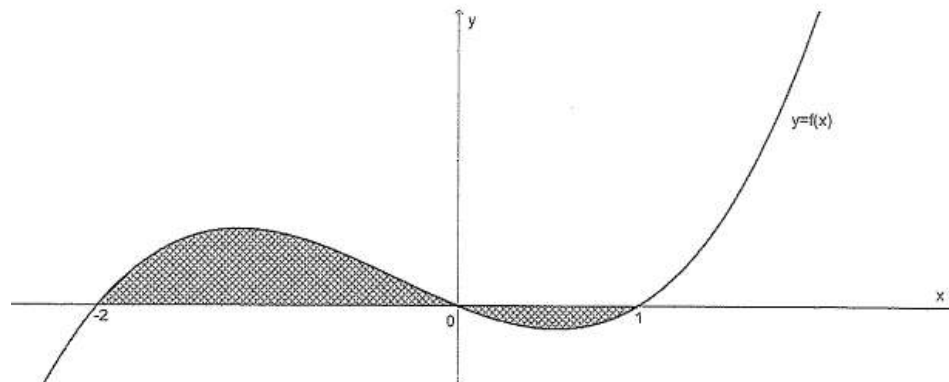
- (A) $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$
- (B) $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$
- (C) $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$
- (D) $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$



$9) (x-4)(x+2) = 0$
 cuts at $x = 4$ and -2
 $\left| \int_0^4 \right| + \int_4^6$ (D)

D7	Girraween	☒	☐
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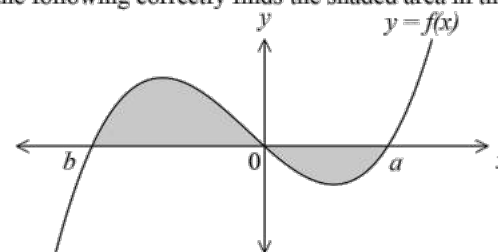
Which would give the value of the shaded area?



- (A) $\int_0^{-2} f(x) dx + \left| \int_0^1 f(x) dx \right|$ (C) $\left| \int_0^{-2} f(x) dx \right| + \int_0^1 f(x) dx$
- (B) $\int_{-2}^0 f(x) dx + \left| \int_0^1 f(x) dx \right|$ (D) $\left| \int_{-2}^0 f(x) dx \right| + \int_0^1 f(x) dx$

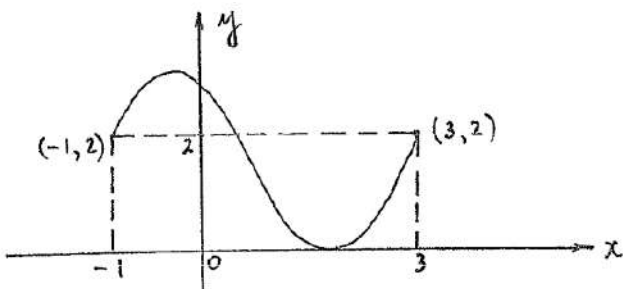
D8	Baulkham Hills	☒	☐
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Which of the following correctly finds the shaded area in this diagram?



- (A) $\int_b^a f(x) dx$ (B) $\left| \int_b^a f(x) dx \right|$
- (C) $\left| \int_0^a f(x) dx \right| + \int_b^0 f(x) dx$ (D) $\int_0^a f(x) dx + \left| \int_b^0 f(x) dx \right|$

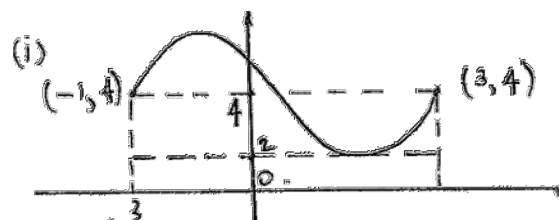
D9	Sydney Grammar	☒	☐
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In the diagram above, the graph of $y = f(x)$ is drawn.

(i) Sketch the graph of $y = f(x) + 2$.

(ii) Given that $\int_{-1}^3 f(x) dx = \frac{15}{2}$, evaluate $\int_{-1}^3 (f(x) + 2) dx$.



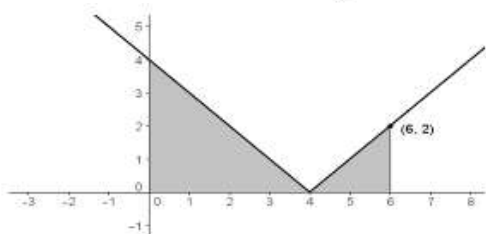
$$(i) \int_{-1}^3 f(x) + 2 dx = \frac{15}{2} + 8 = \frac{31}{2}$$

D10	Caringbah	☒	☐
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Consider the function $f(x) = |x - 4|$

i) Sketch the function.

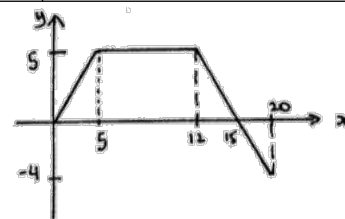
ii) Hence, or otherwise, evaluate $\int_0^6 |x - 4| dx$



$$\int_0^6 |x - 4| dx = \text{area under the curve}$$

$$= \frac{1}{2}(4 \times 4) + \frac{1}{2}(2 \times 2) = 10$$

D11	Ace	☒	☐
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The diagram shows the function $y = f(x)$ between $x = 0$ and $x = 20$.

(i) Evaluate $\int_0^{20} f(x) dx$.

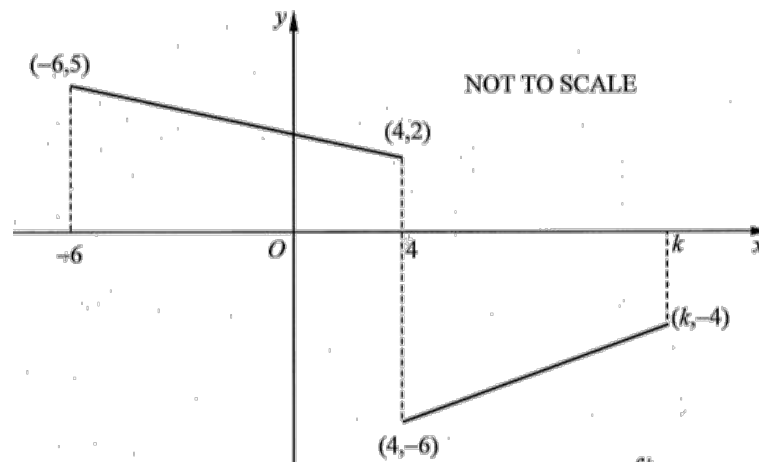
(ii) Is the area between the function $y = f(x)$, the x axis and the lines $x = 0$ and $x = 20$ greater, smaller or equal to $\int_0^{20} f(x) dx$.

Explain fully

$$a) \int_0^{20} f(x) dx = 12 \cdot \frac{1}{2} + 35 + \frac{15}{2} + (-10) = 45$$

ii. The area would be more as the section under the x axis becomes +ve with area increasing the value i.e. Area = $12 \cdot \frac{1}{2} + 35 + \frac{15}{2} + |-10|$

D12	CSSA	☒	☐
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Use the graph above to find the value of k which satisfies $\int_{-6}^k f(x) dx = 0$.

(A) 6

(B) 10

(C) 11

(D) 12

D13	Baulkham Hills	☒	☐
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The area bounded by the graph of $y = f(x)$, the line $x = -2$, the line $x = 4$ and the x -axis, as shaded in the diagram below is A units².

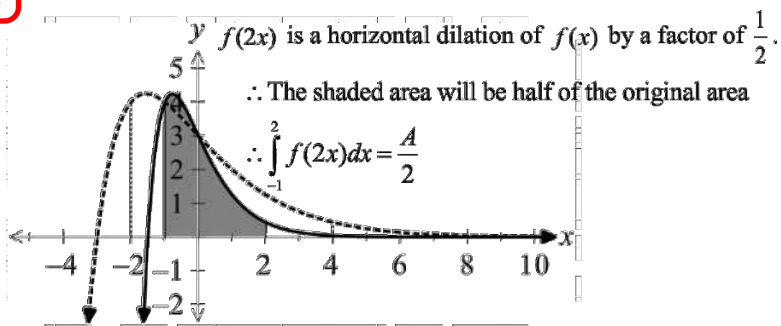
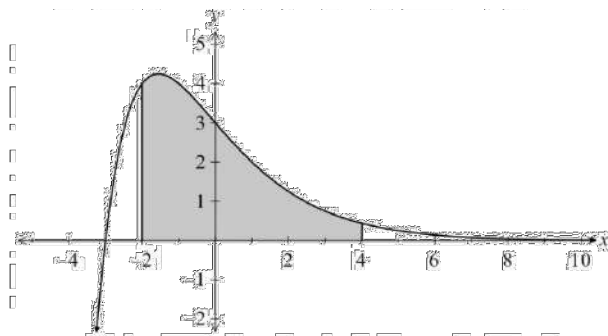
Which of the following integrals below is correct for $f(2x)$?

(A) $\int_{-4}^8 f(2x)dx = 2A$

(B) $\int_{-4}^8 f(2x)dx = A$

(C) $\int_{-1}^2 f(2x)dx = A$

(D) $\int_{-1}^2 f(2x)dx = \frac{A}{2}$



D14	ACE	☒	☐
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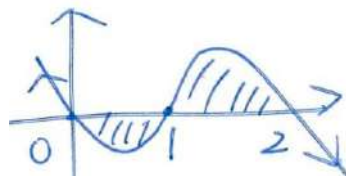
The area enclosed by the function $y = x(x-1)(2-x)$ and the x -axis can be described as:

A. $-\int_0^2 x(x-1)(2-x)dx$

B. $\int_0^1 x(x-1)(2-x)dx - \int_1^2 x(x-1)(2-x)dx$

C. $\int_0^2 x(x-1)(2-x)dx$

D. $-\int_0^1 x(x-1)(2-x)dx + \int_1^2 x(x-1)(2-x)dx$



D15	Hurlstone	☒	☐
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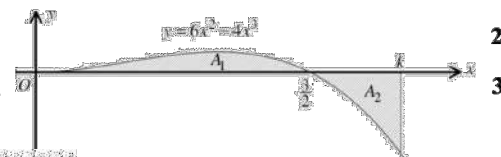
The figure shows the graph of the curve with equation $y = 6x^2 - 4x^3$.

The curve meets the x -axis at the origin and at the point $(\frac{3}{2}, 0)$.

The point $(k, 0)$, $k > \frac{3}{2}$ is such that the area A_1 is equal to the area A_2 .

(a) Show that the area $A_1 = \frac{27}{16}u^2$.

(b) Hence, or otherwise determine the value of k .



$$A_1 = \int_0^{\frac{3}{2}} (6x^2 - 4x^3) dx = \left[2x^3 - x^4 \right]_0^{\frac{3}{2}} = \frac{27}{16}u^2$$

$$\int_{\frac{3}{2}}^k (6x^2 - 4x^3) dx = -\frac{27}{16}$$

$$\left[2x^3 - x^4 \right]_{\frac{3}{2}}^k = -\frac{27}{16}$$

$$2k^3 - k^4 - \frac{27}{4} + \frac{81}{16} = -\frac{27}{16}$$

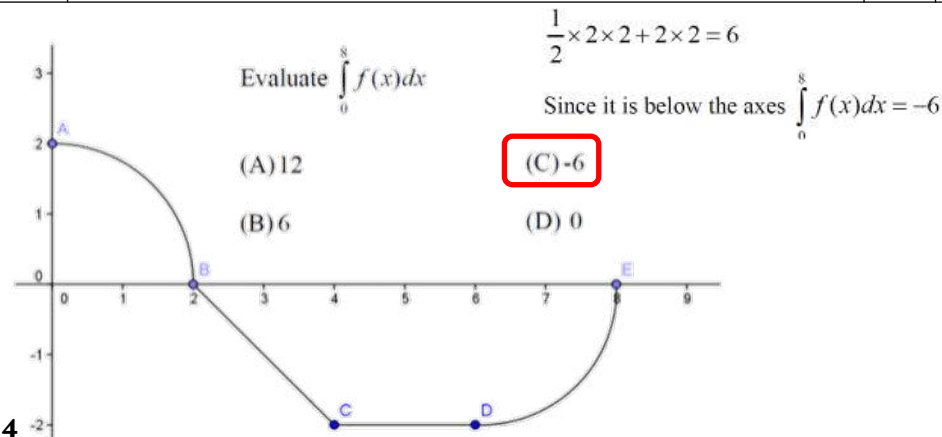
$$2k^3 - k^4 = 0$$

$$k^3(2 - k) = 0$$

$$k = 0 \quad \text{or} \quad k = 2$$

$$\therefore k = 2 \quad (k > \frac{3}{2})$$

D16	Fort Street	☒	☐
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D17	Sydney Girls	☒	☐
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The curve $y = 2\sqrt{x-1}$ is shown below. Find the shaded area.

Find x-intercept
 $2\sqrt{x-1} = 0$
 $\sqrt{x-1} = 0$
 $x-1 = 0$
 $x = 1$
 Area rectangle
 $= 10 \times 6$
 $= 60$

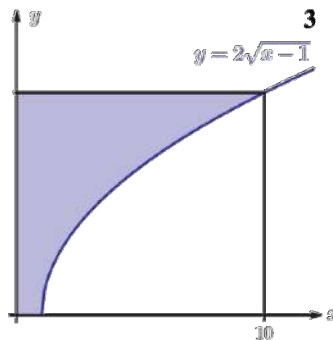
$$A = 60 - 2 \int_1^{10} \sqrt{x-1} \, dx$$

$$= 60 - 2 \int_1^{10} (x-1)^{1/2} \, dx$$

$$= 60 - 2 \left[\frac{2}{3} (x-1)^{3/2} \right]_1^{10}$$

$$= 60 - \frac{4}{3} (27 - 0)$$

$$= 24 \text{ units}^2$$



D18	ACE	☒	☐
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Hence or otherwise, find the value of a , where $a > 0$ such that

$$\int_{-\frac{3}{4}}^a (2|2x+1| - 1) \, dx = 0$$

$$\int_{-0.75}^{-0.25} 2|2x+1| - 1 \, dx = \frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$$

$$\int_{-0.25}^a 2|2x+1| - 1 \, dx = \frac{1}{4}$$

$$\frac{1}{2} \times \left(a + \frac{1}{4}\right) \times (2(2a+1) - 1) = \frac{1}{4}$$

$$\left(a + \frac{1}{4}\right) \times (4a+1) = \frac{1}{2}$$

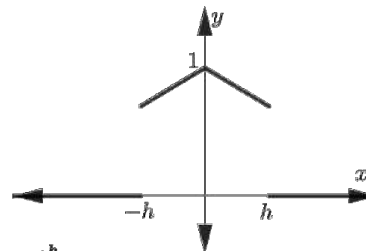
$$4a^2 + 2a + \frac{1}{4} = \frac{1}{2}$$

$$16a^2 + 8a - 1 = 0$$

$$a = \frac{-8 \pm \sqrt{8^2 - 4 \times 16}}{2 \times 16} = -\frac{1}{4} + \frac{\sqrt{2}}{4} \quad (a > 0)$$

D19	St George Girls	☒	☐
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A function and its graph is displayed below, where h is a positive constant.



$$f(x) = \begin{cases} 0 & \text{for } x < -h \\ 1 - \left|\frac{x}{2}\right| & \text{for } -h \leq x \leq h \\ 0 & \text{for } x > h \end{cases}$$

If $\int_{-h}^h f(x) \, dx = 2$, what is the value of h ?

(A) 0.5

(B) 1

(C) 1.5

(D) 2

$$\int_{-h}^h f(x) \, dx = 2 \rightarrow \int_0^h f(x) \, dx = 1$$

$$Area = \frac{1}{2} \times (1 + 1 - \frac{1}{2}) = 1$$

$$\frac{1}{2} \left(2 - \frac{1}{2}\right) = 1$$

$$h - \frac{1}{4} = 1$$

$$4h - 1 = 4$$

$$4h = 5$$

$$h = \frac{5}{4}$$

D20	Sydney Boys	☒	☐
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A graph of the function $y = f(x)$ is shown below. It is given that:

• Area A is equal to 3 units²

• $\int_0^3 f(x) \, dx = 4$

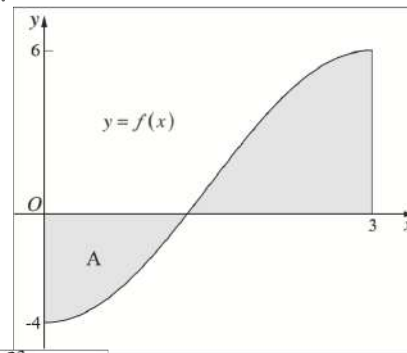
What is the area that the graph of $y = f(x)$ makes with the y -axis?

A. 4

B. 10

C. 14

D. 20



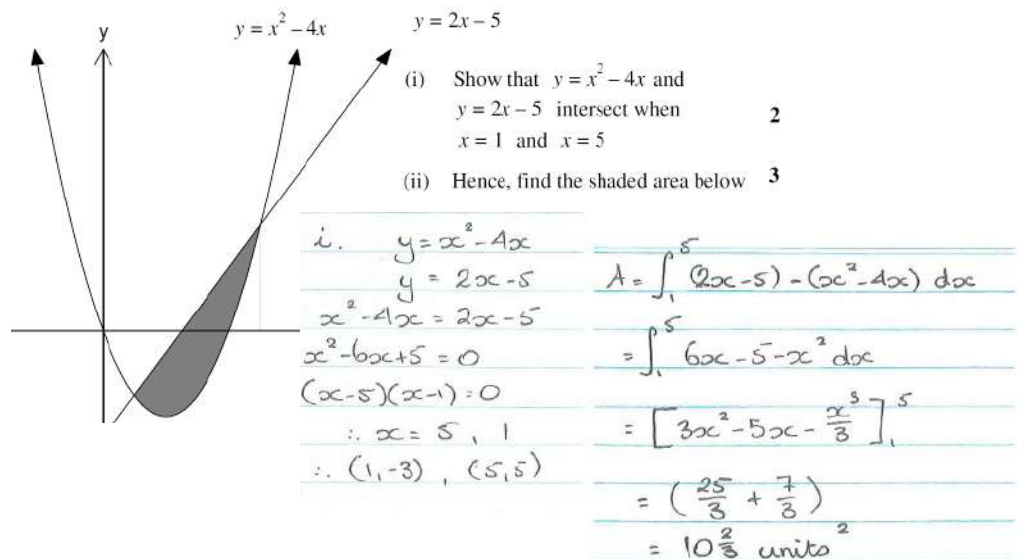
$$\int_0^3 f(x) \, dx = \int_0^b f(x) \, dx + \int_b^3 f(x) \, dx$$

$$\therefore 4 = -4 + \int_b^3 f(x) \, dx$$

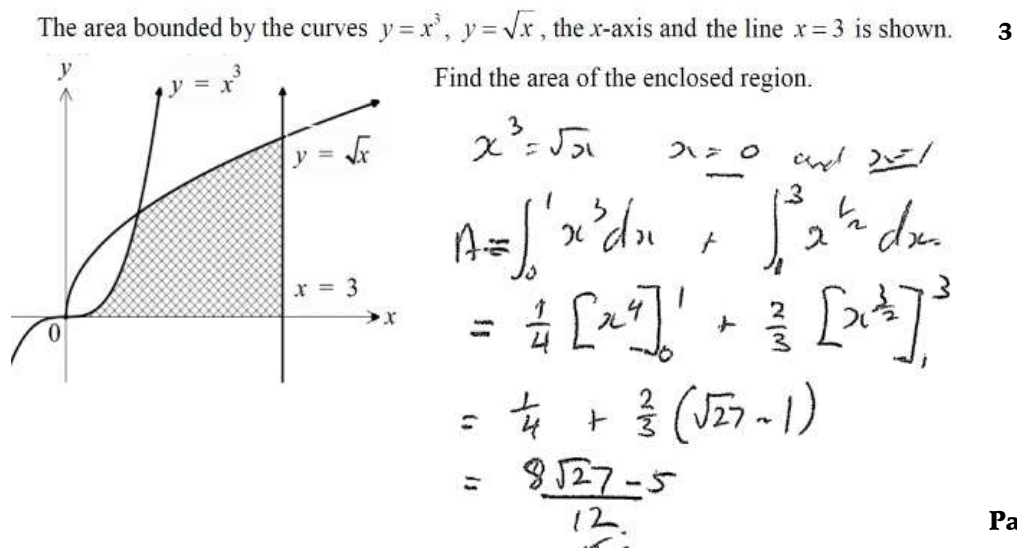
$$\therefore \int_b^3 f(x) \, dx = 8$$

Area of rectangle = $6 \times 3 = 18$
 $\therefore$ remaining area of rectangle = $18 - 7 = 11$
 $\therefore$ Area next to y -axis = $11 + 3 = 14$
 Area of rectangle = $6 \times 3 = 18$
 Area next to y -axis = $11 + 3 = 14$

E1	Barker	☒	☐
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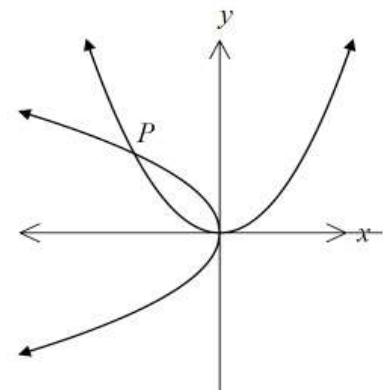
E2	Callaghan	☒	☐
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E3	Abbotsleigh	☒	☐
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The curves $y = x^2$ and $y^2 = -x$ are shown in the diagram below. They intersect at the origin and at the point P .

- (i) Show that P has the coordinates $(-1, 1)$. 1
- (ii) The area bounded by the two curves is rotated about the y -axis. Determine the volume of the solid generated. 2

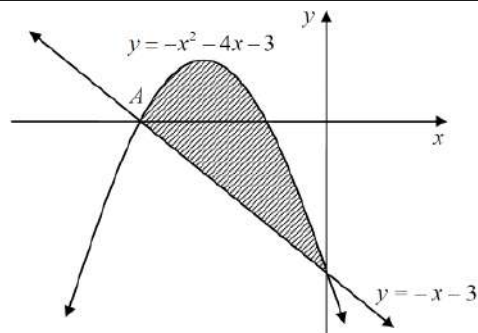


(c) Point P is the intersection of the two curves.

- (i) $y = x^2$ (1)
 $y^2 = -x$ (2)
 Substitute x^2 for y into eqn(2).
 $(x^2)^2 = -x$
 $x^4 + x = 0$
 $x(x^3 + 1) = 0$
 Therefore $x = 0$ or $x = -1$.
 Substitute these values into eqn (1).
 $x = 0$ then $y = 0^2 = 0$, $x = -1$ then $y = (-1)^2 = 1$
 Clearly from the diagram point P is $(-1, 1)$

- (c) (ii) Now $y = x^2$ and $(y^2)^2 = (-x)^2$ or $y^4 = x^2$
 $V = \pi \int_0^1 x^2 dy$
 $= \pi \int_0^1 (y - y^4) dy$
 $= \pi \left[\frac{1}{2} y^2 - \frac{1}{5} y^5 \right]_0^1$
 $= \pi \left(\frac{1}{2} - \frac{1}{5} \right)$
 $= \frac{3\pi}{10}$

E4	Baulkham Hills	☒	☐
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The diagram shows the sketch of the parabola $y = -x^2 - 4x - 3$ and a line $y = -x - 3$

- Find the x coordinate of A .
- Find the area of the shaded region bounded by the line and the parabola.

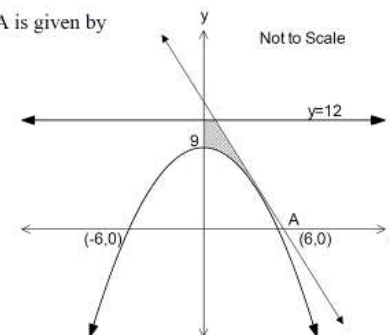
$$\begin{aligned} \text{b) } -x^2 - 4x - 3 &= -x - 3 \\ x^2 + 3x &= 0 \\ x(x+3) &= 0 \\ x &= 0 \text{ or } x = -3 \\ \therefore x \text{ coordinate of } A &\text{ is } -3 \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{(ii) } A &= \int_{-3}^0 [-x^2 - 4x - 3] - [-x - 3] dx \\ &= \int_{-3}^0 (-x^2 - 3x) dx \\ &= \left[-\frac{1}{3}x^3 - \frac{3}{2}x^2 \right]_{-3}^0 \\ &= -\frac{1}{3}(0)^2 - \frac{3}{2}(0)^2 - \left(-\frac{1}{3}(-3)^3 - \frac{3}{2}(-3)^2 \right) \\ &= 4.5 \text{ units}^2 \quad \textcircled{2} \end{aligned}$$

E5	Abbotsleigh	☒	☐
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The graph of $y = 9 - \frac{x^2}{4}$ is drawn below. The tangent to the curve at $A(6,0)$ is also drawn

- Show that the equation of the tangent to the curve at A is given by $y = -3x + 18$.
- What are the co-ordinates of the focus of the parabola $y = 9 - \frac{x^2}{4}$?
- Find the area of the shaded region.



1
2

$$\text{a) } y = 9 - \frac{x^2}{4}$$

$$\text{at } x = 6 \quad y = -3 \quad \textcircled{1}$$

$$\text{Target } y - 0 = -3(x - 6)$$

$$y = -3x + 18 \quad \textcircled{1}$$

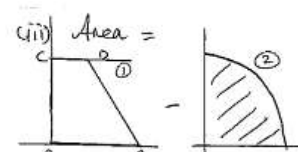
$$\text{(iii) } y = 9 - \frac{x^2}{4}$$

$$\frac{x^2}{4} = -y + 9$$

$$x^2 = -4(y - 9) \quad \textcircled{1}$$

$$\therefore 4a = 4 \quad a = 1$$

$$\text{Focus } (0, 8) \quad \textcircled{1}$$



$$\text{at } D \quad y = 12 \quad y = -3x + 18$$

$$12 = -3x + 18$$

$$x = 2$$

$$\therefore \text{Area of trap.} = \frac{1}{2} \cdot 12(2+6)$$

$$= 48 \text{ units}^2 \quad \textcircled{1}$$

$$\text{Area 2} \Rightarrow \text{make } x \text{ the subject}$$

$$x^2 = -4y + 36$$

$$x = \sqrt{36 - 4y}$$

$$= 2(9 - y)^{\frac{1}{2}} \quad \textcircled{1}$$

$$\begin{aligned} \therefore \text{Area} &= 2 \int_0^9 (9 - y)^{\frac{1}{2}} dy \\ &= 2 \left[\frac{2}{3} (9 - y)^{\frac{3}{2}} \right]_0^9 \quad \textcircled{1} \\ &= -\frac{4}{3} \left[\sqrt{9 - y}^3 \right]_0^9 \\ &= -\frac{4}{3} [0 - 27] \\ &= \frac{108}{3} = 36 \end{aligned}$$

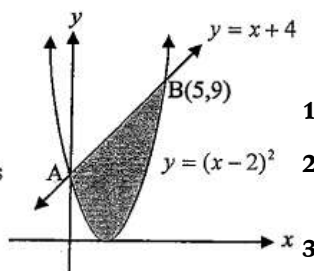
$$\therefore \text{Shaded area} = 48 - 36$$

$$= 12 \text{ units}^2$$

E6	Cranbrook	☒	☐
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The graphs of $y = (x-2)^2$ and $y = x+4$ intersect at the point A and the point $B(5,9)$.

- Show that the point A lies on the y -axis.
- Write down the two inequalities whose intersection describes the shaded area shown in the diagram above.
- Find the area of the shaded region bounded by the graphs of $y = (x-2)^2$ and $y = x+4$.



$$(x-2)^2 = x+4$$

$$x^2 - 4x + 4 = x+4$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0$$

$$x=0, x=5$$

$$y=4, y=9$$

$\therefore A$ is $(0,4)$ which is on y axis as x coord is zero.

$y \leq x+4$ and $y \geq (x-2)^2$

$\textcircled{1}$ for both correct

$$A = \int_0^5 (x+4) - (x-2)^2 dx$$

$$= \int_0^5 (x+4 - (x^2 - 4x + 4)) dx$$

$$= \int_0^5 (5x - x^2) dx$$

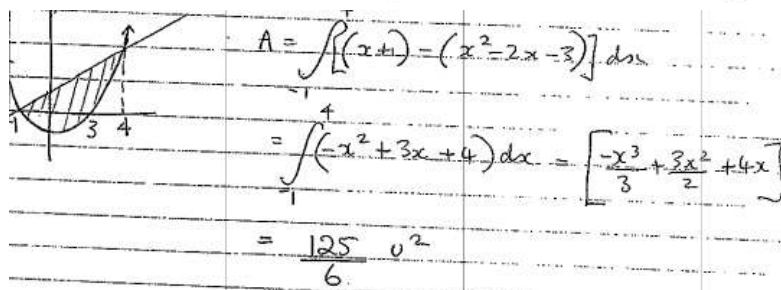
$$= \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^5$$

$$= 62\frac{1}{2} - \frac{125}{3}$$

$$= 20\frac{5}{6} \text{ units}^2$$

E7	ACE	☒	☐
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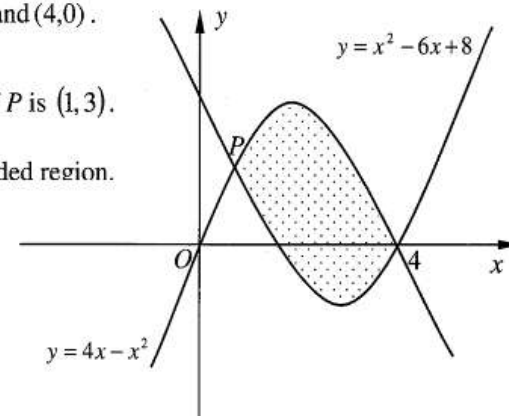
Find the area enclosed between the curve $y = x^2 - 2x - 3$ and the line $y = x+1$ 3



E8	Abbotsleigh	☒	☐
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The graphs of the functions $y = x^2 - 6x + 8$ and $y = 4x - x^2$ are shown in the diagram below. They intersect at P and $(4,0)$.

- Show that the coordinates of P is $(1,3)$.
- Calculate the area of the shaded region.



(b) (i) (1 mark)

Sample answer:

On substitution into $y = x^2 - 6x + 8$, when $x=1$,
 $y = 1^2 - 6 + 8 = 3$

On substitution into $y = 4x - x^2$, when $x=1$,
 $y = 4 - 1 = 3$

$\therefore$ The coordinate of P is $(1, 3)$

(b) (ii) (3 marks)

Sample answer:

$$\text{Area} = \int_1^4 ((4x - x^2) - (x^2 - 6x + 8)) dx$$

$$= \int_1^4 (-2x^2 + 10x - 8) dx$$

$$= -2 \int_1^4 (x^2 - 5x + 4) dx$$

$$= -2 \left[\frac{x^3}{3} - \frac{5x^2}{2} + 4x \right]_1^4$$

$$= -2 \left[\left(\frac{64}{3} - \frac{80}{2} + 16 \right) - \left(\frac{1}{3} - \frac{5}{2} + 4 \right) \right]$$

$$= -2 \left[-\frac{8}{3} - \frac{11}{6} \right]$$

$$= -2 \times -\frac{9}{2}$$

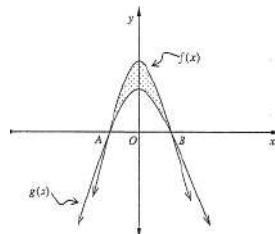
$$= 9 \text{ units}^2$$

E9	CSSA	☒	☐
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The graphs of $f(x) = (5+x)(5-x)$ and $g(x) = \frac{2}{5}(5+x)(5-x)$ intersect at points A and B as shown in the diagram above.

(i) Show that the area of the shaded region is given by

$$A = \frac{6}{5} \int_0^5 (25 - x^2) dx.$$



(ii) Hence, or otherwise, evaluate the area of the shaded region.

(d) (i) (1 mark)

Sample answer

$$\begin{aligned} A &= 2 \int_0^5 \left[(5+x)(5-x) - \frac{2}{5}(5+x)(5-x) \right] dx \\ &= 2 \int_0^5 \frac{3}{5}(5+x)(5-x) dx \\ &= \frac{6}{5} \int_0^5 (5+x)(5-x) dx \end{aligned}$$

(d) (ii) (2 marks)

Sample answer

$$\begin{aligned} A &= \frac{6}{5} \int_0^5 (25 - x^2) dx \\ &= \frac{6}{5} \left[25x - \frac{x^3}{3} \right]_0^5 \\ &= \frac{6}{5} \left[\left(125 - \frac{125}{3} \right) - (0 - 0) \right] \\ &= \frac{250}{3} \times \frac{6}{5} \\ &= 100 \text{ u}^2 \end{aligned}$$

E10	CSSA	☒	☐
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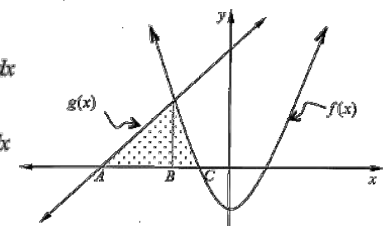
The shaded region is best described by:

(A) $\int_A^B g(x) dx + \int_B^C f(x) dx$

(B) $\int_A^B g(x) dx - \int_B^C f(x) dx$

(C) $\int_A^C g(x) + f(x) dx$

(D) $\int_A^C f(x) - g(x) dx$



E11	Fort Street	☒	☐
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Find the area bounded by the line $y = 3x + 1$ and the parabola with equation

$$y = 2x^2 + 4x - 5$$

(c) $y = 3x + 1, y = 2x^2 + 4x - 5$

$$3x + 1 = 2x^2 + 4x - 5$$

$$0 = 2x^2 + x - 6$$

$$= 2x^2 + 4x - 3x - 6$$

$$= 2x(x + 2) - 3(x + 2)$$

$$= (x + 2)(2x - 3)$$

$$\therefore x = -2, x = 1 \frac{1}{2} \quad \bullet$$

$$A = \int_{-2}^{1 \frac{1}{2}} (3x + 1) - (2x^2 + 4x - 5) dx \quad \bullet$$

$$= \int_{-2}^{1 \frac{1}{2}} -2x^2 - x + 6 dx$$

$$= \left[-\frac{2}{3}x^3 - \frac{1}{2}x^2 + 6x \right]_{-2}^{1 \frac{1}{2}}$$

$$= -\frac{2}{3} \left(\frac{3}{2} \right)^3 - \frac{1}{2} \left(\frac{3}{2} \right)^2 + 6 \left(\frac{3}{2} \right) - \left[-\frac{2}{3}(-2)^3 - \frac{1}{2}(-2)^2 + 6(-2) \right]$$

$$= \frac{343}{24} \text{ units}^2 \quad \bullet$$

E12	CSSA	☒	☐
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The graph of the function $f(x) = \frac{9-x^2}{3}$ is shown below.

The graph intersects the x -axis and the y -axis at the points A and B respectively.

The tangent to the graph at point P is parallel to the line AB .

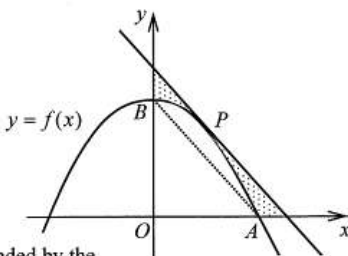
The coordinates of B are $(0, 3)$.

(i) Find the coordinates of the point A .

(ii) Show that the coordinates of point P are $(\frac{1}{2}, 2\frac{1}{4})$.

(iii) Find the equation of the tangent at point P .

(iv) The shaded region shown in the diagram above is bounded by the curve $y = f(x)$, the tangent at P , the x -axis and y -axis.



Show that the area of this shaded region is $\frac{33}{32}$ units².

Question 15 (15 marks)

(a) (i) (1 mark)

Sample answer:

Let $y = 0 \Rightarrow x = 3$ ($x > 0$) $\therefore A(3, 0)$

(a) (ii) (3 marks)

Sample answer:

Gradient of the line $AB = \frac{0-3}{3-0} = -1$

$\therefore$ gradient of the tangent at the point P is -1

$$f(x) = \frac{9-x^2}{3} = 3 - \frac{1}{3}x^2$$

$$\therefore f'(x) = -\frac{2}{3}x \Rightarrow -1 = -\frac{2}{3}x$$

$$\therefore x = \frac{1}{2} \text{ and } y = 2\frac{1}{4} \therefore P \text{ is } (\frac{1}{2}, 2\frac{1}{4})$$

(a) (iii) (1 mark)

Sample answer:

$$y - 2\frac{1}{4} = -1(x - \frac{1}{2}) \Rightarrow y - \frac{9}{4} = -x + \frac{3}{2}$$

$$\therefore 4x + 4y - 15 = 0$$

(a) (iv) (3 marks)

Sample answer:

$$4x + 4y - 15 = 0 \text{ and if } y = 0, x = \frac{3}{4}$$

$\therefore$ the x -intercept of the tangent is $(\frac{3}{4}, 0)$

$$\text{and if } x = 0, y = \frac{3}{4}$$

$\therefore$ the y -intercept of the tangent is $(0, \frac{3}{4})$

Area of triangle enclosed by tangent & axes

$$= \frac{1}{2} \times \frac{3}{4} \times \frac{3}{4} = \frac{225}{32} u^2$$

$$\text{Now, area under the curve} = \int_0^{\frac{3}{4}} (3 - \frac{1}{3}x^2) dx$$

$$= [3x - \frac{x^3}{9}]_0^{\frac{3}{4}}$$

$$= (9 - \frac{27}{64}) - 0 = 6\frac{57}{64} u^2$$

$$\therefore \text{The shaded area} = \frac{225}{32} - 6\frac{57}{64} = \frac{33}{32} u^2$$

E13	Girraween	☒	☐
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(a) Solve $-x^2 + 13x - 36 = 0$.

(b) Find the equation of the tangent to the parabola $y = -x^2 + 13x - 36$ at the point where $x = 6$.

(c) Draw a diagram showing the parabola and the tangent. Shade the region bounded by the parabola, the tangent and the x -axis.

(d) Find the area shaded in the diagram above.

$$(a) -x^2 + 13x - 36 = 0$$

$$\therefore x^2 - 13x + 36 = 0$$

$$(x-4)(x-9) = 0$$

$$\therefore x = 4 \text{ or } x = 9 \quad \# \quad (2)$$

$$(b) y = -x^2 + 13x - 36$$

$$\text{at } x=6; y = -6^2 + 13 \times 6 - 36 = 6$$

$$\frac{dy}{dx} = -2x + 13$$

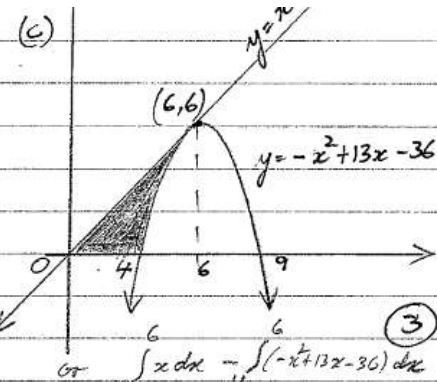
$$\text{at } x=6, \frac{dy}{dx} = -2 \times 6 + 13 = 1$$

equation of tangent is

$$y - 6 = 1(x - 6)$$

$$y - 6 = x - 6$$

$$\therefore y = x \quad \# \quad (3)$$



(d) Area shaded

$$= \frac{1}{2} \times 6 \times 6 - \int_4^9 (-x^2 + 13x - 36) dx$$

$$= 18 - \left[-\frac{x^3}{3} + \frac{13x^2}{2} - 36x \right]_4^9$$

$$= 18 - \left[-\frac{9^3}{3} + \frac{13 \times 9^2}{2} - 36 \times 9 \right] + \left[-\frac{4^3}{3} + \frac{13 \times 4^2}{2} - 36 \times 4 \right]$$

$$= 18 - [-54] - [61\frac{2}{3}]$$

$$= 18 + 54 - 61\frac{2}{3}$$

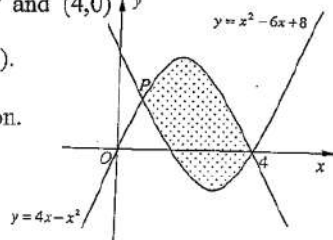
$$= 10\frac{2}{3} \text{ units}^2 \quad \# \quad (4)$$

E14	Girraween	☒	☐
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The graphs of the functions $y = 4x - x^2$ and $y = x^2 - 6x + 8$ are shown in the diagram below. They intersect at P and $(4,0)$.

(i) Show that the coordinates of P is $(1,3)$.

(ii) Calculate the area of the shaded region.



$$\begin{aligned} x^2 - 6x + 8 &= 4x - x^2 \\ 2x^2 - 10x + 8 &= 0 \Rightarrow x^2 - 5x + 4 = 0 \\ (x-4)(x-1) &= 0 \therefore x = 1, 4 \end{aligned}$$

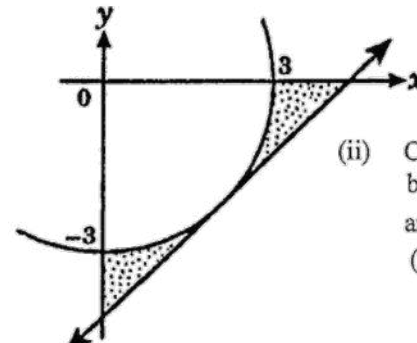
$$\therefore y = (3, 0)$$

$$\therefore P = (1, 3) \quad (1)$$

$$\begin{aligned} A &= \int_1^4 (4x - x^2) - (x^2 - 6x + 8) dx \\ &= \int_1^4 (4x - x^2 - x^2 + 6x - 8) dx \\ &= \int_1^4 (-2x^2 + 10x - 8) dx = \left[-\frac{2x^3}{3} + 5x^2 - 8x \right]_1^4 \\ &= \left(-\frac{2}{3} \cdot 64 + 80 - 32 \right) - \left(-\frac{2}{3} + 5 - 8 \right) \\ &= 9 \text{ u}^2 \quad (3) \end{aligned}$$

E15	Girraween	☒	☐
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(i) Prove that the line $3x - 4y = 15$ is a tangent to the circle $x^2 + y^2 = 9$.

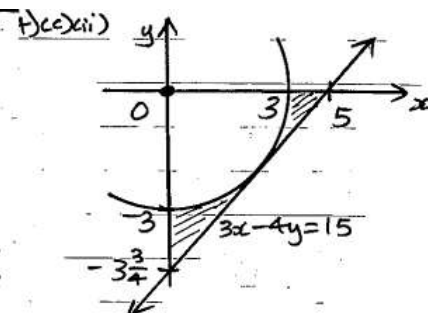


(ii) Calculate the exact area in the fourth quadrant between the tangent $3x - 4y = 15$ and the circle $x^2 + y^2 = 9$ (as shown by the shaded area in the diagram above).

(c)(ii) $3x - 4y = 15$ crosses:

x-axis: when $y=0$: $3x=15$
 $x=5$
x-intercept is $(5, 0)$

y-axis: when $x=0$: $-4y=15$
 $y=-\frac{15}{4}$
 $y=-3\frac{3}{4}$

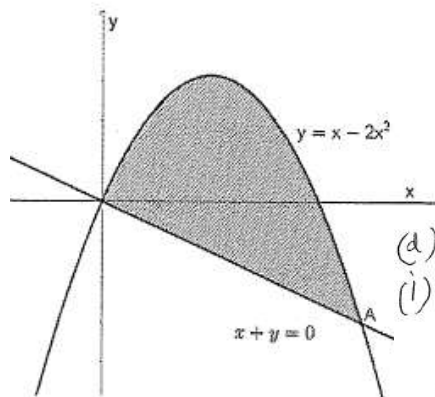


Shaded Area

$$\begin{aligned} &= (\text{Area triangle}) - (\text{Area Quadrant}) \\ &= \left(\frac{1}{2} \times 5 \times \frac{15}{4} \right) - \left(\frac{1}{4} \times \pi \times 3^2 \right) \\ &= \frac{75}{8} - \frac{9\pi}{4} \\ &= \frac{75 - 18\pi}{8} \\ &= \frac{3(25 - 6\pi)}{8} \text{ square units} \quad (2m) \end{aligned}$$

E16	Girraween	☒	☐
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The graph below shows the area enclosed by the parabola $y = x - 2x^2$ and the line $x + y = 0$. The parabola and the line intersect at the origin and point A.



(i) Find the coordinates of point A. 1

(ii) Find the value of the shaded area. 3

$$\begin{aligned}
 (d) \quad & \begin{cases} y = -x \\ y = x - 2x^2 \end{cases} \quad (11) \text{ Area} = \int (\text{top graph} - \text{bottom graph}) \\
 (i) \quad & -x = x - 2x^2 \\
 & 0 = 2x - 2x^2 \\
 & 0 = 2x(1 - x) \\
 & \begin{cases} x = 0 \\ y = 0 \end{cases} \quad \begin{cases} x = 1 \\ y = -1 \end{cases} \\
 & \text{So } A = (1, -1) \\
 & \text{Area} = \int_0^1 (x - 2x^2) - (-x) dx \\
 & = \int_0^1 (2x - 2x^2) dx \\
 & = \left[x^2 - \frac{2}{3}x^3 \right]_0^1 \\
 & = \left(1 - \frac{2}{3} \right) - 0 \\
 & = \frac{1}{3} \text{ units}^2
 \end{aligned}$$

E17	Killara	☒	☐
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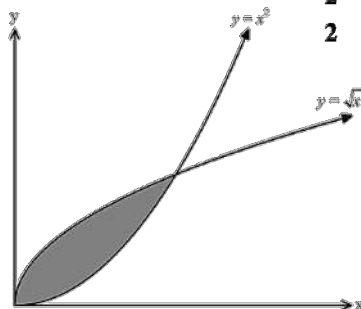
The graph of $y = x^2$ and $y = \sqrt{x}$ are shown below for $x \geq 0$.

(a) Find the points of intersection between the two graphs.

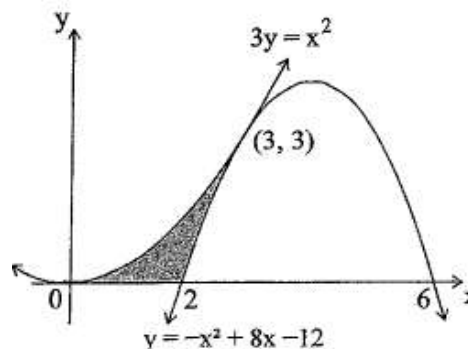
(b) Hence, find the area of the shaded region.

$$\begin{aligned}
 (a) \quad & x^2 = \sqrt{x} \\
 & x^4 = x \\
 & x^4 - x = 0 \\
 & x(x^3 - 1) = 0 \\
 & \therefore x = 0, 1 \\
 & \therefore y = 0, 1
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int_0^1 \sqrt{x} - x^2 dx \\
 & = \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3} \right]_0^1 \\
 & = \frac{2}{3} - \frac{1}{3} \\
 & = \frac{1}{3}
 \end{aligned}$$



E18	Gosford	☒	☐
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The graphs of $3y = x^2$ and $y = -x^2 + 8x - 12$ are shown on the same set of axes.

These curves meet at (3, 3) as shown.

The region enclosed by these curves in the first quadrant has been shaded.

Calculate the area of this region.

$$\begin{aligned}
 & \text{The area under the curve } y = \frac{x^2}{3} \text{ between } x=0 \text{ and } x=3 \text{ is } A_1. \\
 & A_1 = \int_0^3 \frac{x^2}{3} dx = \left[\frac{x^3}{9} \right]_0^3 = 3 \text{ units}^2 \\
 & \text{The area under the curve } y = -x^2 + 8x - 12 \text{ between } x=2 \text{ and } x=3 \text{ is } A_2. \\
 & A_2 = \int_2^3 (-x^2 + 8x - 12) dx \\
 & = \left[-\frac{x^3}{3} + 4x^2 - 12x \right]_2^3 \\
 & = \left(-9 + 36 - 36 \right) - \left(-\frac{8}{3} + 16 - 24 \right) \\
 & = 1\frac{2}{3} \text{ units}^2. \\
 & \text{The shaded area can be found by subtracting } A_2 \text{ from } A_1. \\
 & \therefore A_1 - A_2 = 3 - 1\frac{2}{3} = 1\frac{1}{3} \text{ units}^2. \quad (3 \text{ marks})
 \end{aligned}$$

E19	Hornsby	☒	☐
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- (i) Find the points of intersection of the parabola $y = x^2 + 3x - 5$ and the line $y = 2x + 1$. **2**
(ii) Hence find the area enclosed between the parabola $y = x^2 + 3x - 5$ and the line $y = 2x + 1$. **2**

(b)
(i)
 $y = x^2 + 3x - 5 \dots (1)$
 $y = 2x + 1 \dots (2)$
Equating (1) and (2)
 $2x + 1 = x^2 + 3x - 5$
 $0 = x^2 + x - 6$
 $0 = (x + 3)(x - 2)$
 $x = -3, x = 2$

When $x = 2, y = 5$ and $x = -3, y = -5$

The points of intersection are $(2, 5)$ and $(-3, -5)$

(11)

$$A = \int_{-3}^2 ((2x + 1) - (x^2 + 3x - 5)) dx$$

$$= \int_{-3}^2 (-x^2 - x + 6) dx$$

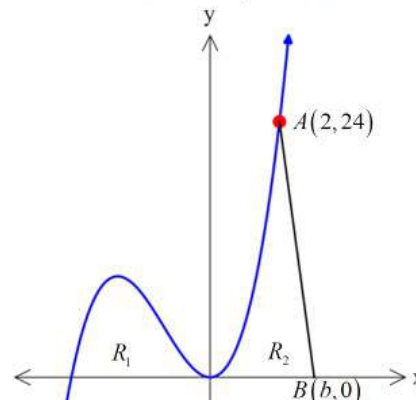
$$= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 6x \right]_{-3}^2$$

$$= \left(\frac{-8}{3} - 2 + 12 \right) - \left(\frac{27}{3} - \frac{9}{2} - 18 \right)$$

$$= \frac{125}{6} \text{ units}^2$$

E20	Hornsby	☒	☐
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The sketch below shows part of the curve with equation $y = x^2(x + 4)$.



The finite region R_1 is bounded by the curve and the negative x -axis.

The finite region R_2 is bounded by the curve, the positive x -axis and AB , where $A = (2, 24)$ and $B(b, 0)$ where $b > 2$.

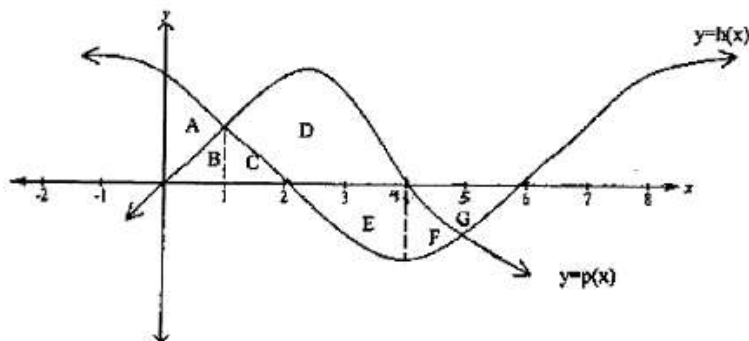
- (i) Show that the area of R_1 is $\frac{64}{3}$ square units. **2**
(ii) If the area of the regions R_1 and R_2 are equal, find the exact value of b . **3**

(b)
(i)
Area of $R_1 = \int_{-4}^0 x^2(x + 4) dx$
 $= \int_{-4}^0 (x^3 + 4x^2) dx$
 $= \left[\frac{x^4}{4} + \frac{4x^3}{3} \right]_{-4}^0$
 $= (0) - \left(64 - \frac{256}{3} \right)$
 $= \frac{64}{3} \text{ units}^2$

(ii)
 $R_2 = \frac{64}{3}$
 $\frac{64}{3} = \int_0^2 (x^3 + 4x^2) dx + \frac{1}{2} \times 24 \times (b - 2)$
 $= \left[\frac{x^4}{4} + \frac{4x^3}{3} \right]_0^2 + 12(b - 2)$
 $= \frac{2^4}{4} + \frac{4(2)^3}{3} + 12b - 24$
 $= 12b - \frac{28}{3}$
 $12b = \frac{64}{3} + \frac{28}{3}$
 $12b = \frac{92}{3}$
 $b = \frac{23}{9}$

E21	ACE	☒	☐
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The two curves $y = p(x)$ and $y = h(x)$ are sketched below. Different areas enclosed by the curves and the axes are labelled A to G.



The integrals below represent the sum of which areas.

(i) $\int_0^4 p(x) dx$

(ii) $\int_1^5 p(x) - h(x) dx$

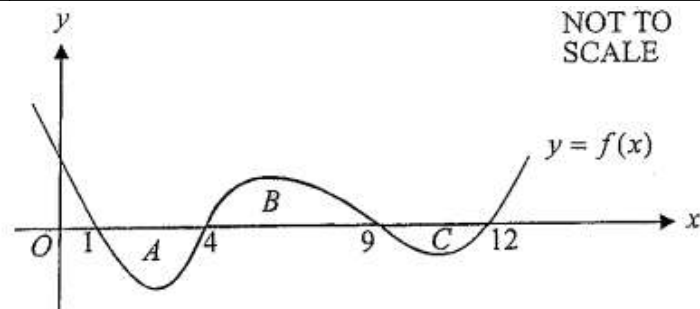
(iii) Give an integral which would define the area denoted by letter G

$$i) \int_0^4 p(x) dx = B + C + D$$

$$ii) \int_1^5 p(x) - h(x) dx = D + E + F$$

$$iii) \left| \int_4^5 p(x) dx \right| + \left| \int_5^8 h(x) dx \right|$$

E22	Ace	☒	☐
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NOT TO SCALE

3

The graph of the function f is shown in the diagram above. The shaded area A is equal in area to the shaded area B.

Write an alternative, equivalent expression for $\int_1^{12} f(x) dx$ in terms of the integral of $f(x)$.

(Do not attempt to evaluate any integrals.)

$$\int_1^{12} f(x) dx = \text{area B} - \text{area A} - \text{area C}$$

because $\int_1^4 f(x) dx$ and $\int_9^{12} f(x) dx$ are both negative.

Since area A = area B,

$$\begin{aligned} \int_1^{12} f(x) dx &= -\text{area C} \\ &= \int_9^{12} f(x) dx \end{aligned}$$

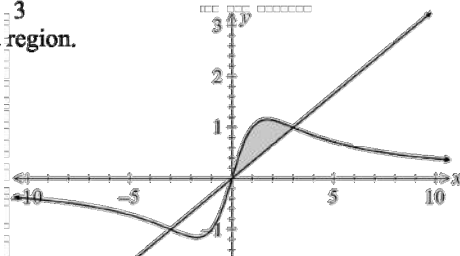
E23	Baulkham Hills	☒	☐
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The curve $y = \frac{4x}{x^2+3}$ intersects the line $y = \frac{x}{3}$ at the points $(0,0)$, $(3,1)$ and $(-3,1)$. The region bounded by the curve $y = \frac{4x}{x^2+3}$ and the line $y = \frac{x}{3}$ between $x=0$ and $x=3$ is shaded in the diagram below. Find the exact area of the shaded region.

$$A = \int_0^3 \frac{4x}{x^2+3} - \frac{x}{3} dx = 2 \ln 12 - \frac{9}{6} - 2 \ln 3$$

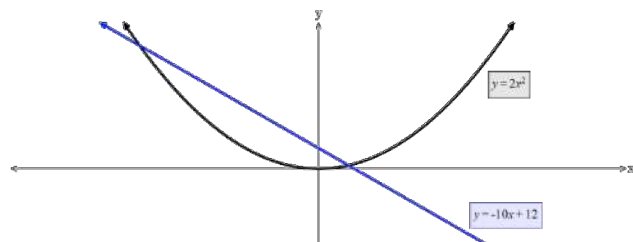
$$= 2 \int_0^3 \frac{2x}{x^2+3} dx - \int_0^3 \frac{x}{3} dx = 2(\ln 12 - \ln 3) - \frac{3}{2}$$

$$= 2 \ln 4 - \frac{3}{2}$$



E24	Girraween	☒	☐
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Consider the curve $y = 2x^2$ and the line $y = -10x + 12$. Find the area enclosed between the two functions. Give answer to 1 decimal place.



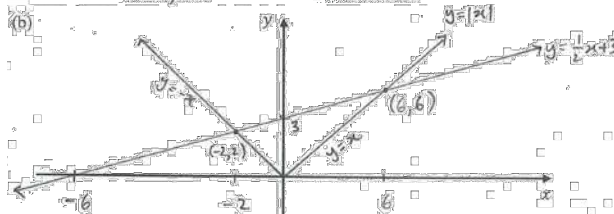
P.O.I $2x^2 = -10x + 12$
 $2x^2 + 10x - 12 = 0$
 $x^2 + 5x - 6 = 0$
 $(x-1)(x+6) = 0$
 $\therefore x = 1 \text{ and } -6$

$$A = \int_{-6}^1 -10x + 12 - 2x^2 dx = \left[-5x^2 + 12x - \frac{2x^3}{3} \right]_{-6}^1 = \frac{343}{3} = 114.3 \text{ u}^2$$

E25	Penrith	☒	☐
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- (a) Solve $|x| = \frac{1}{2}x + 3$ 2
- (b) On the axes below draw a neat sketch showing the graphs of $y = |x|$ and $y = \frac{1}{2}x + 3$. Show the x and y intercepts and any points of intersection. 2
- (c) An area between $y = |x|$ and $y = \frac{1}{2}x + 3$ is bounded by $x = -2$ and $x = k$, where $k > 0$. Find the value of k such that the area is 10 square units. 4

(a) $|x| = \frac{1}{2}x + 3$
 $2|x| = x + 6$
 $x = 6$
 $x = -2$



(c) $A = \int_{-2}^0 (-x) dx + \int_0^k \left(\frac{1}{2}x + 3 - x \right) dx$
 $A = \int_{-2}^0 -x dx + \int_0^k \left(-\frac{1}{2}x + 3 \right) dx$
 $A = \left[-\frac{1}{2}x^2 \right]_{-2}^0 + \left[-\frac{1}{4}x^2 + 3x \right]_0^k$
 $A = 2 + \left(-\frac{1}{4}k^2 + 3k \right)$
 $10 = 2 - \frac{1}{4}k^2 + 3k$
 $8 = -\frac{1}{4}k^2 + 3k$
 $32 = -k^2 + 12k$
 $k^2 - 12k + 32 = 0$
 $(k-4)(k-8) = 0$
 $k = 4 \text{ or } 8$

$7 = \int_{-2}^k \left(\frac{1}{2}x + 3 - x \right) dx$
 $7 = \int_{-2}^k \left(-\frac{1}{2}x + 3 \right) dx$
 $7 = \left[-\frac{1}{4}x^2 + 3x \right]_{-2}^k$
 $7 = -\frac{1}{4}k^2 + 3k - 2$
 $9 = -\frac{1}{4}k^2 + 3k$
 $36 = -k^2 + 12k$
 $k^2 - 12k + 36 = 0$
 $(k-6)^2 = 0$
 $k = 6$
 $k = 6 = 2\sqrt{2}$
 $\text{But } k = 6$
 $k = 6 = 2\sqrt{2} \text{ only}$

E26	Penrith	☒	☐
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Aneka and Brian are investigating areas A and B respectively on either side of the graph of $y = \ln x$ as shown on the diagram below:

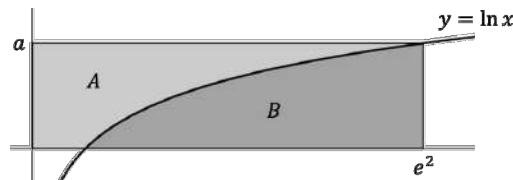
- Find the value of a .
- Brian uses the trapezoidal rule with 3 function values to estimate the area B . He approximates all of the values used in his working out to 2 decimal places where necessary. What is his estimate for the area of B ?

(a) $a = \ln e^2 = 2$

(b) $h = \frac{e^2 - 1}{2}$

x	1	$\frac{e^2+1}{2}$	e^2
y	0	1.43	2

$$\text{Area} = \frac{e^2 - 1}{2(2)} (0 + 1.43) + \frac{e^2 - 1}{2(2)} (1.43 + 2) = 7.77 \text{ (2 d.p.)}$$



E27	Penrith	☒	☐
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Find the area bounded between the graphs of $y = \sqrt{x-1}$ and $y = x-1$.

3

$$\sqrt{x-1} = x-1$$

$$x-1 = x^2 - 2x + 1$$

$$x^2 - 3x + 2 = 0$$

$$(x-1)(x-2) = 0$$

$$x=1, x=2$$

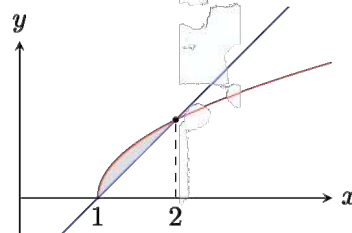
$$A = \int_1^2 \sqrt{x-1} - (x-1) dx$$

$$A = \int_1^2 (x-1)^{1/2} - x + 1 dx$$

$$A = \left[\frac{2}{3}(x-1)^{3/2} - \frac{x^2}{2} + x \right]_1^2$$

$$A = \frac{2}{3} - 2 + 2 - \left(-\frac{1}{2} + 1 \right)$$

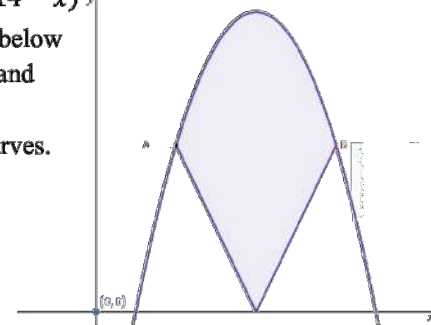
$$A = \frac{1}{6}$$



E28	Sydney Tech	☒	☐
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The curves $y = \frac{5}{4}|x-8|$ and $y = \frac{1}{12}(3x-6)(14-x)$ intersect at two points as shown in the diagram below

- Show that the point A has coordinates (4,5) and hence write down the coordinates of point B.
- Calculate the shaded area between the two curves.



i. When $x=4$ $y = \frac{5}{4}|4-8| = 5$ & $y = \frac{1}{12}(3 \times 4 - 6)(14 - 4) = \frac{1}{12}(6)(10) = 5$

so meet @ A (4,5) with symmetry

$$B = (12, 5)$$

ii. $A = \int_4^{12} \frac{1}{12}(3x-6)(14-x) dx - 2 \times \frac{1}{2} \times 5 \times 4$

$$= \frac{1}{12} \int_4^{12} 42x - 3x^2 - 84 + 6x dx - 20$$

$$= \frac{1}{12} \int_4^{12} 48x - 3x^2 - 84 dx - 20$$

$$= \frac{1}{12} [24x^2 - x^3 - 84x]_4^{12} - 20$$

$$= \frac{1}{12} [24 \times 12^2 - 12^3 - 84 \times 12 - (24 \times 16 - 4^3 - 84 \times 4)] - 20$$

$$= 41 \frac{1}{3} u^2$$